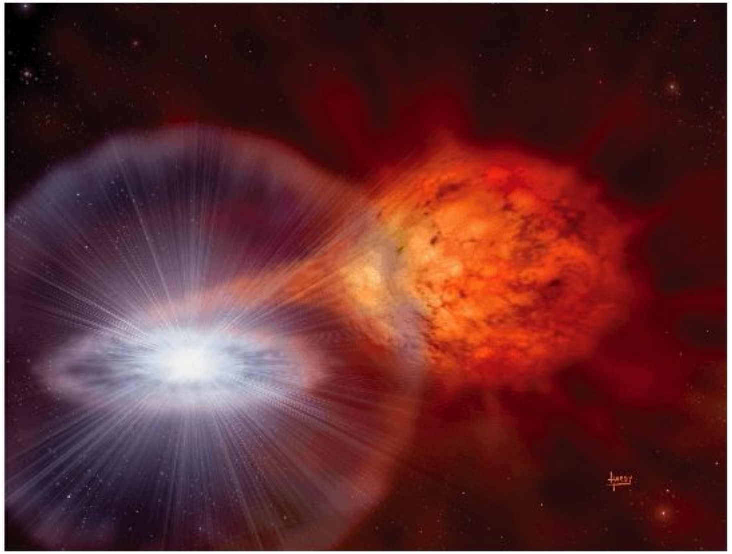
Unveiling the Intrinsic Type Ia Supernova Mass Step

Matt Grayling

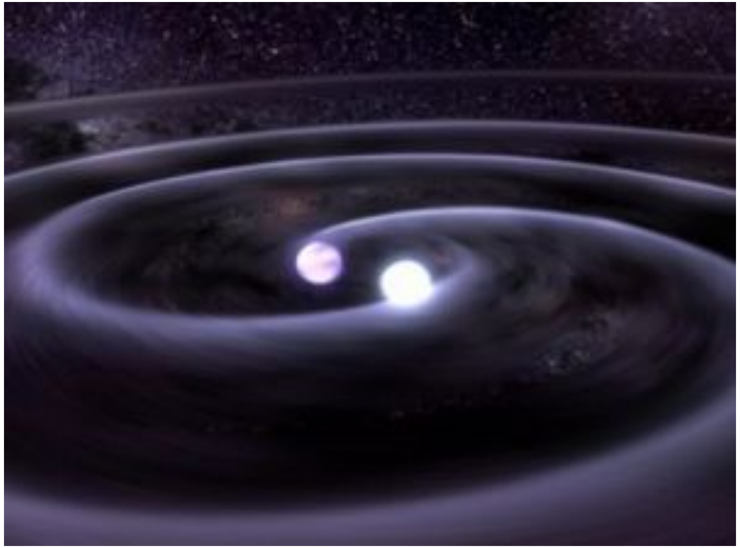
Type Ia Supernovae

Single Degenerate



Credit: David A. Hardy, © David A. Hardy/astroasrt.org

Double Degenerate



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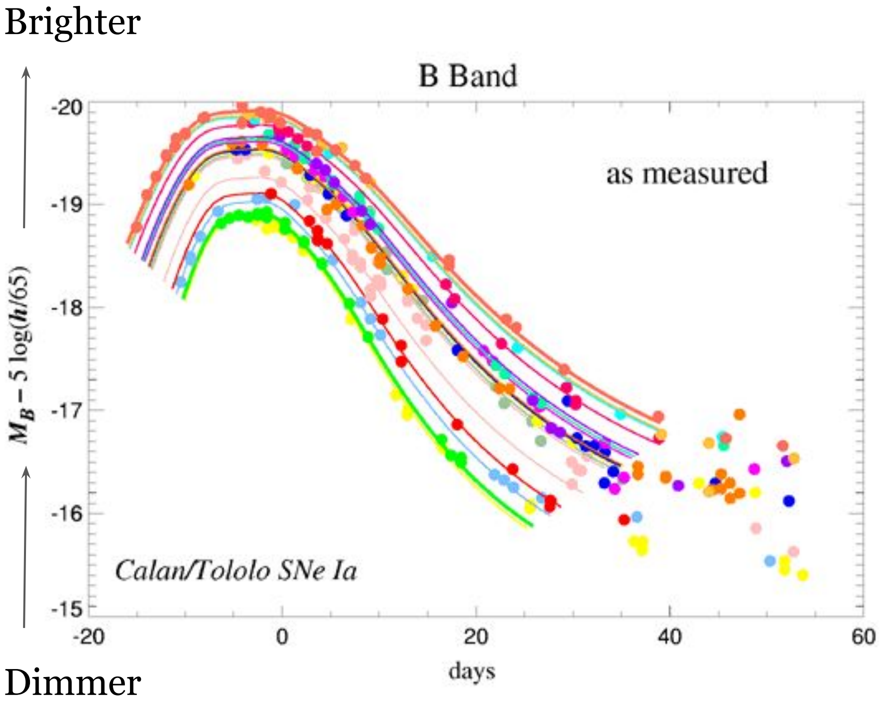
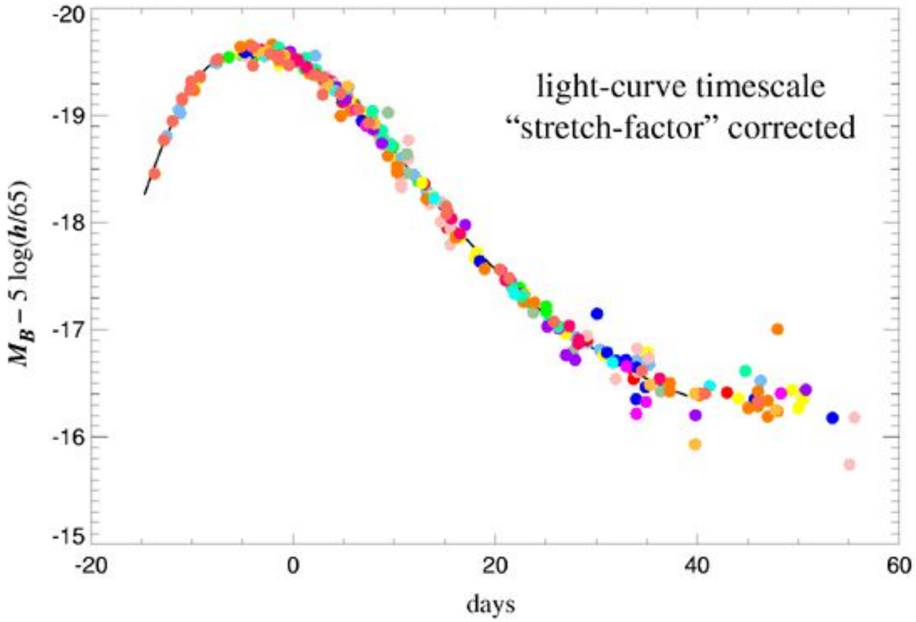
Standard(isable) Candles



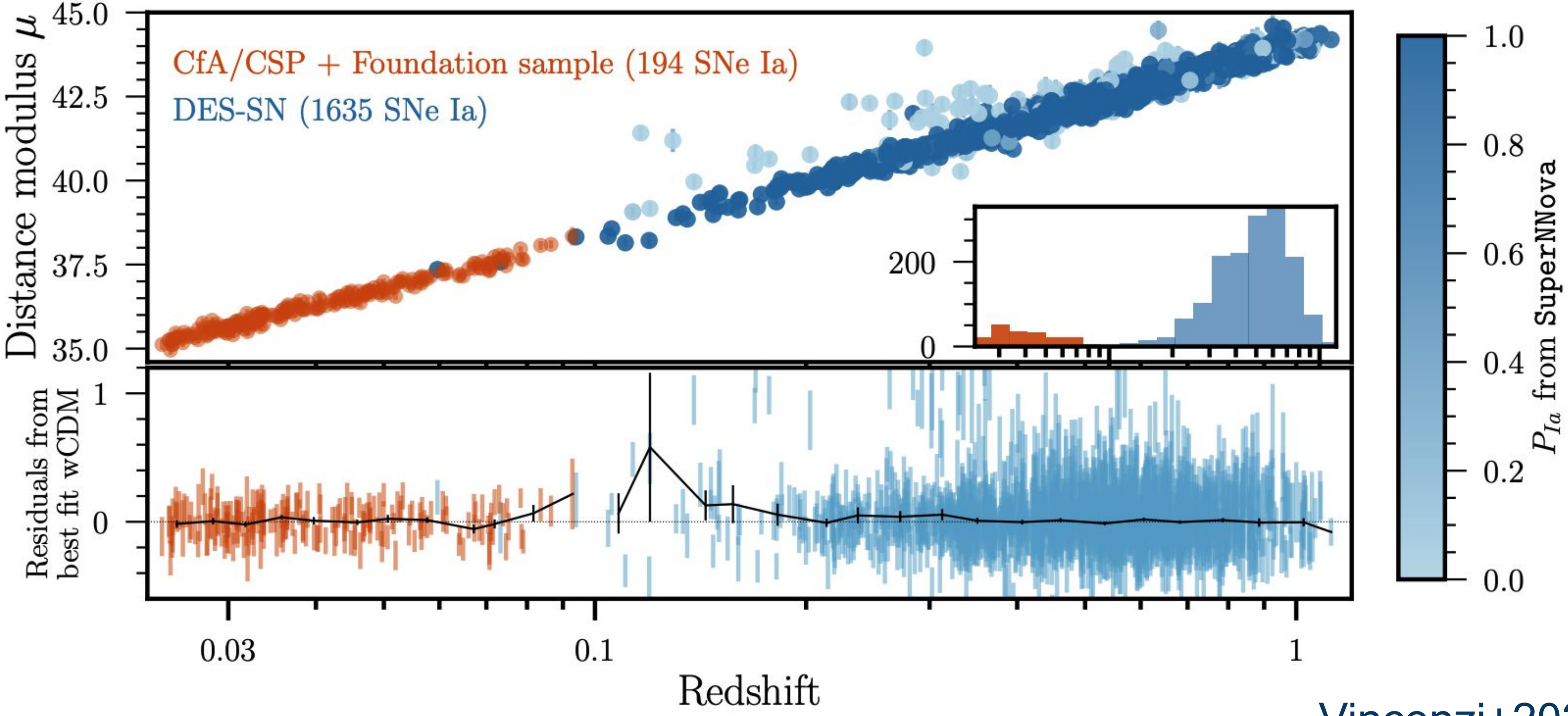


From Supernova Cosmology Project

Calculating Distance

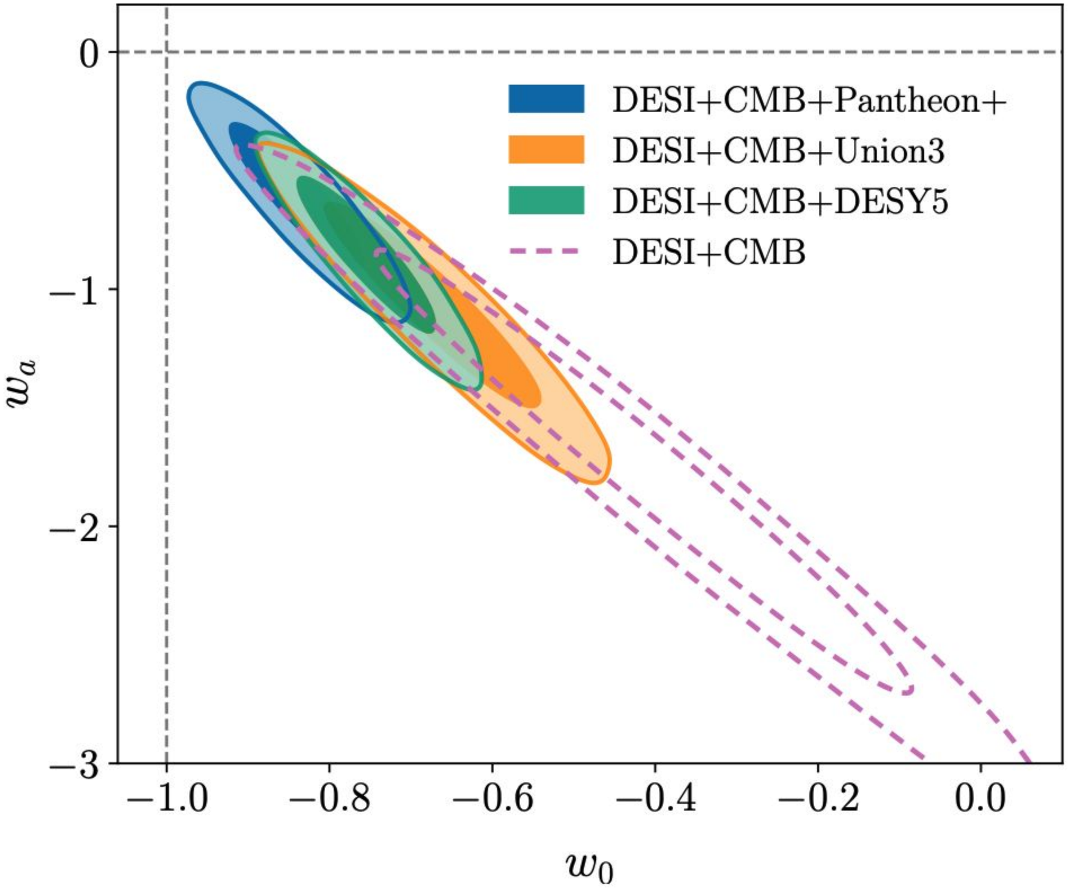
$$\mu^s = m_B^s - M + \alpha x_1^s - \beta c^s$$

The Hubble Diagram



Vincenzi+2024

Crisis in Cosmology?



DESI Collaboration 2025

Supernova cosmology is easy, right?

The Impact of Dust

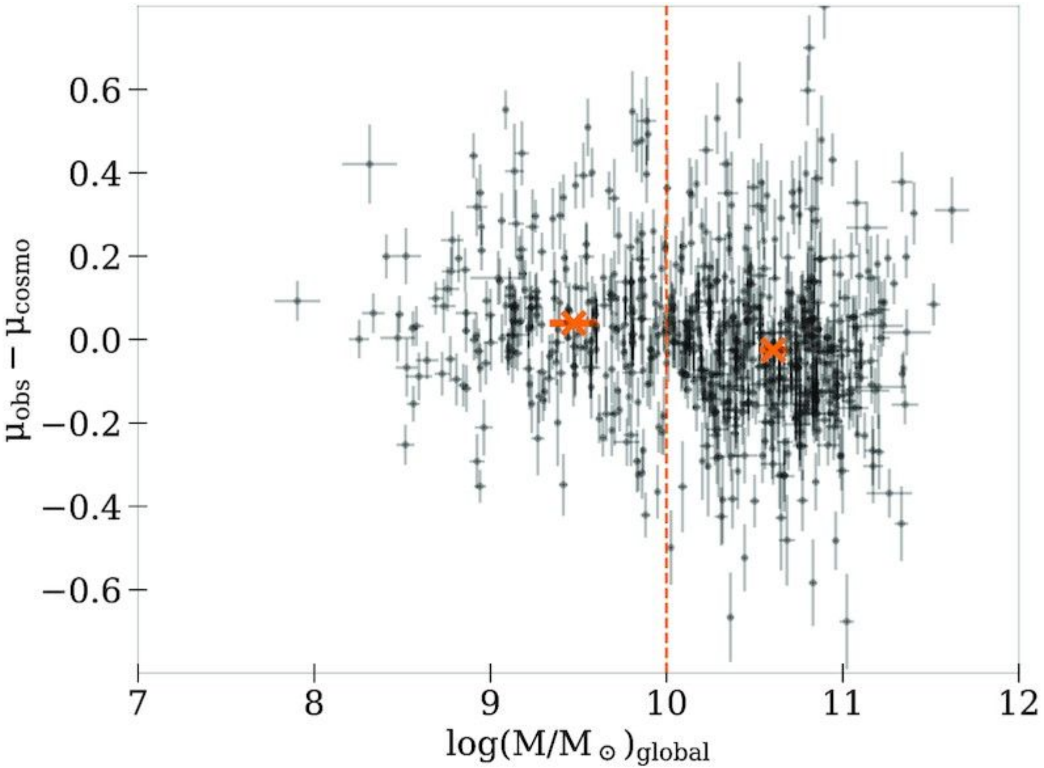
$$\mu^s = m_B^s - M + \alpha x_1^s - \beta c^s$$

$$c^s = \color{blue}{c_{\mathrm{int}}^s} + \color{red}{E(B-V)^s}$$

- Colour - one parameter for two separate effects
- Correctly handling dust is key to uncover intrinsic SN properties and accurately estimate distances

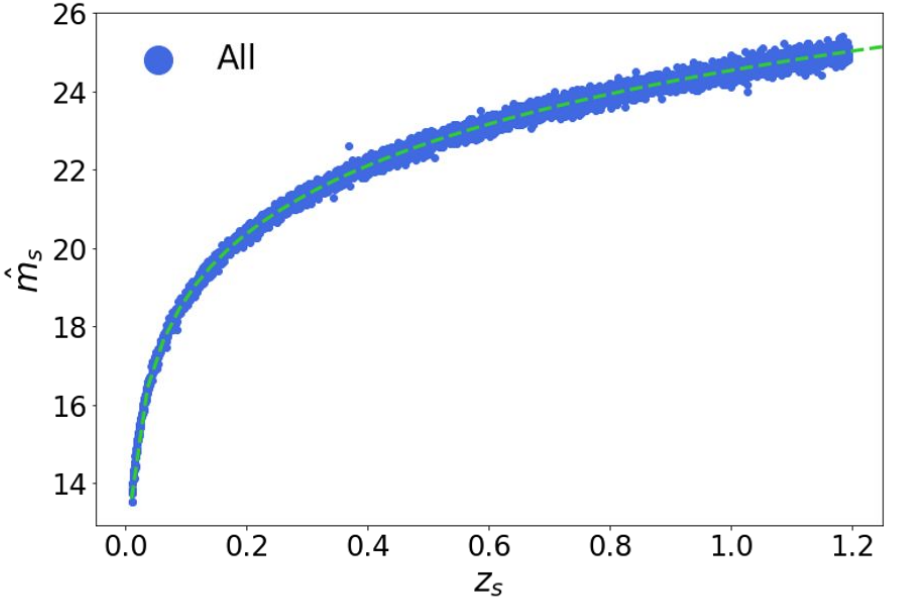
The Mass Step

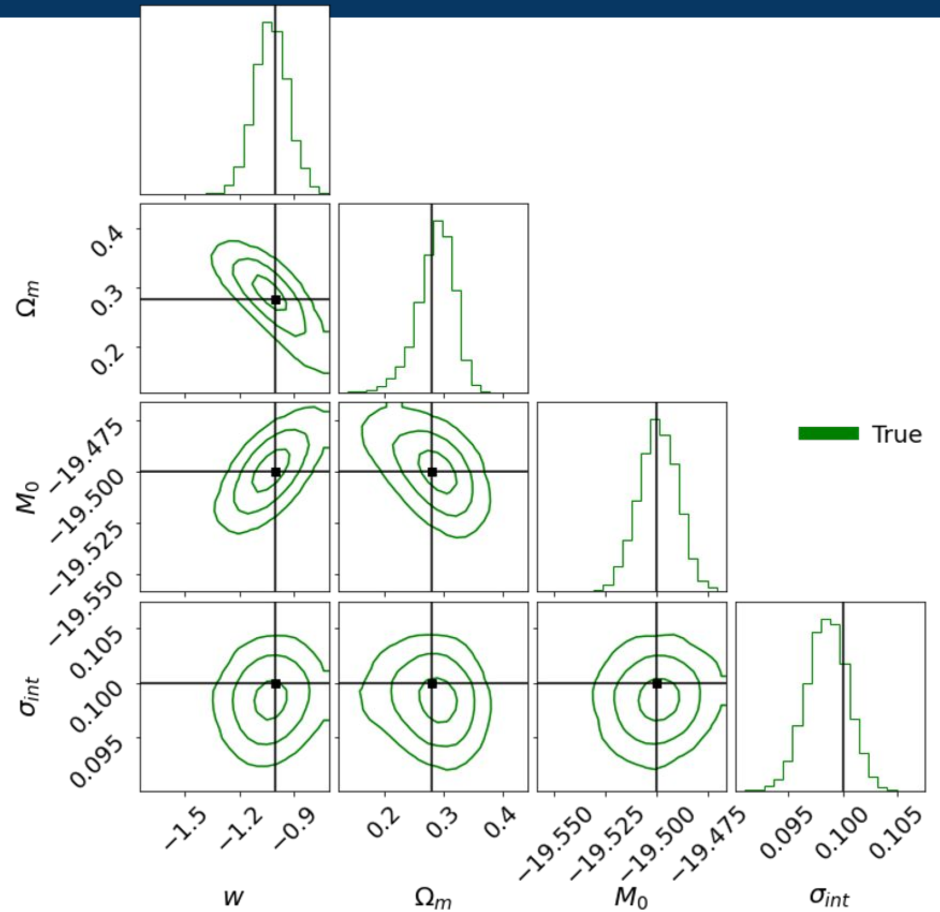
- SN Ia properties depend on environment
- Just different dust properties?
  - SN populations identical, just appear different?
- Or intrinsic differences?
  - Interesting physical implications, different progenitor channels?



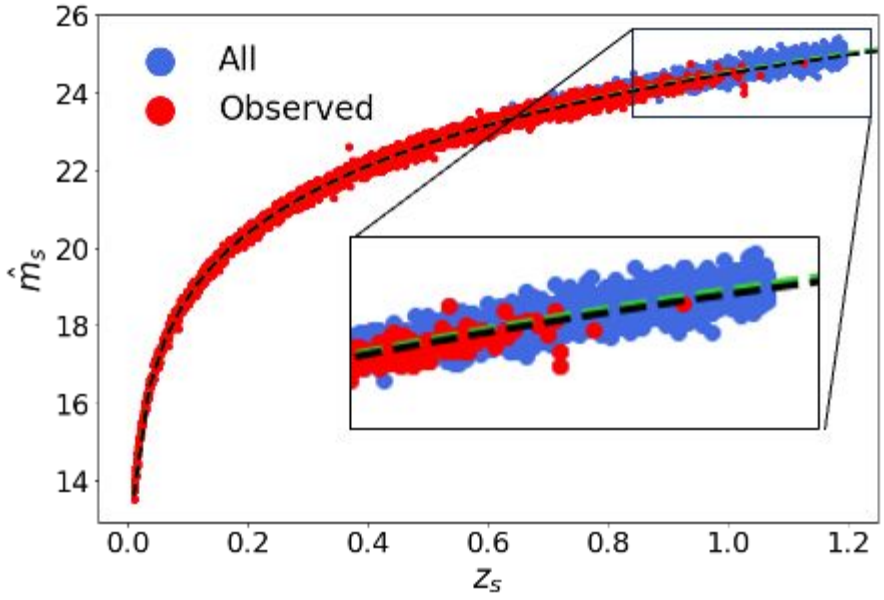
Kelsey+2023

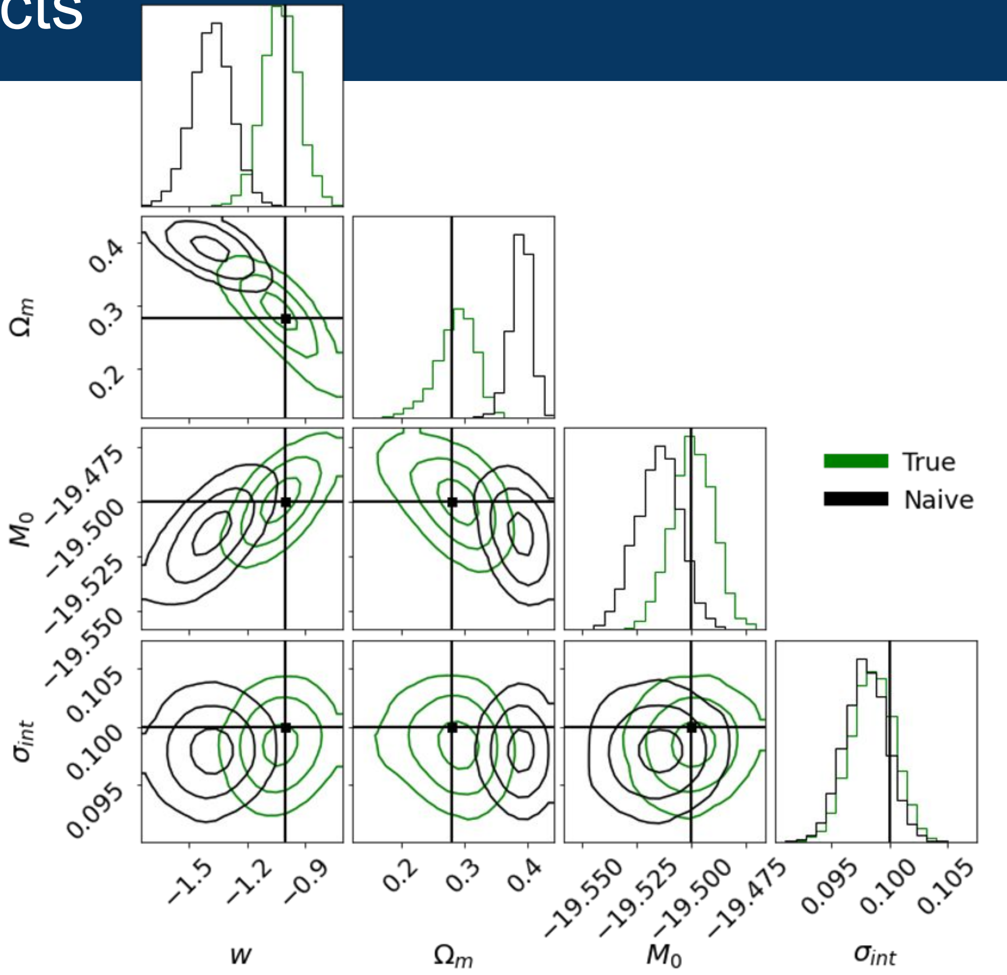
The Impact of Selection Effects



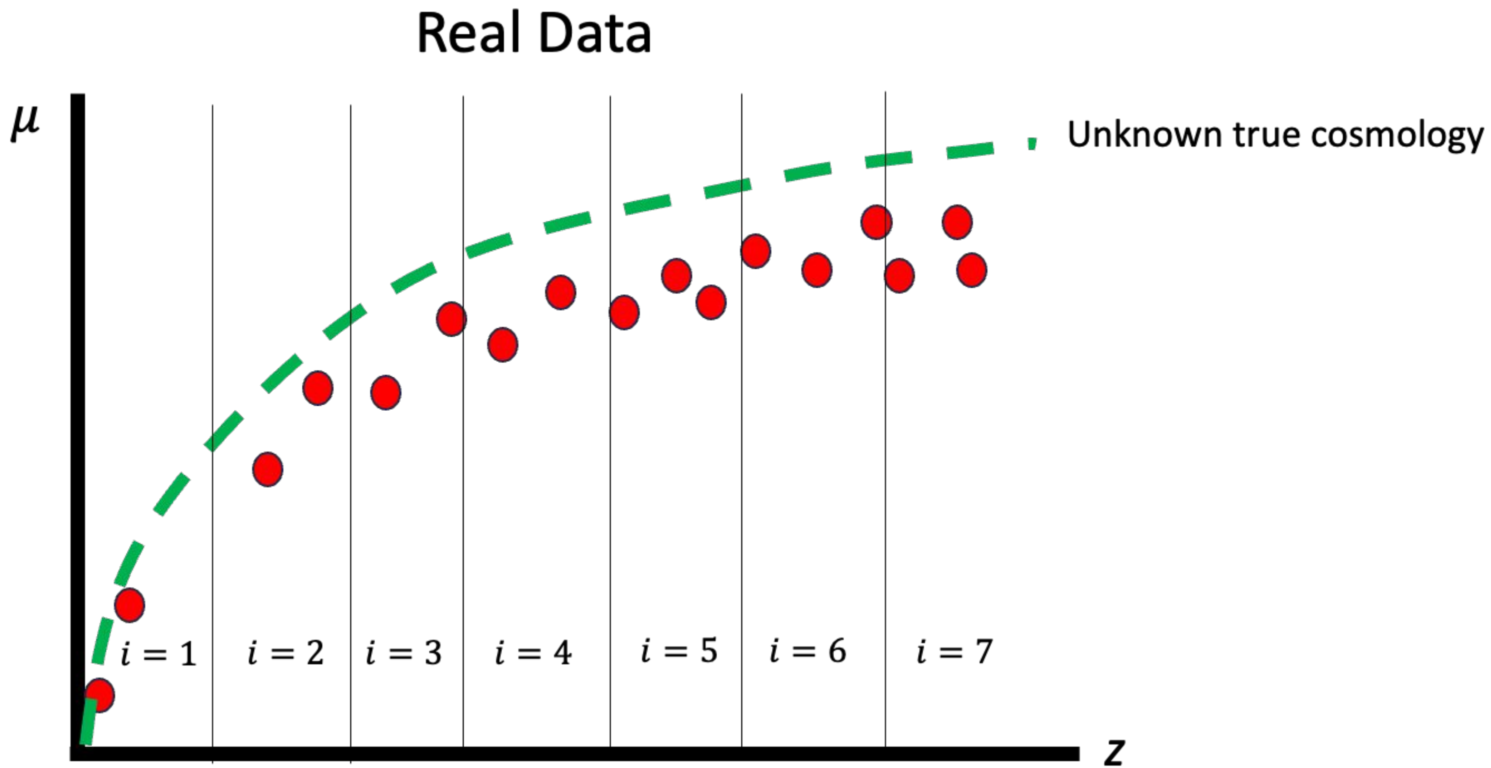


The Impact of Selection Effects

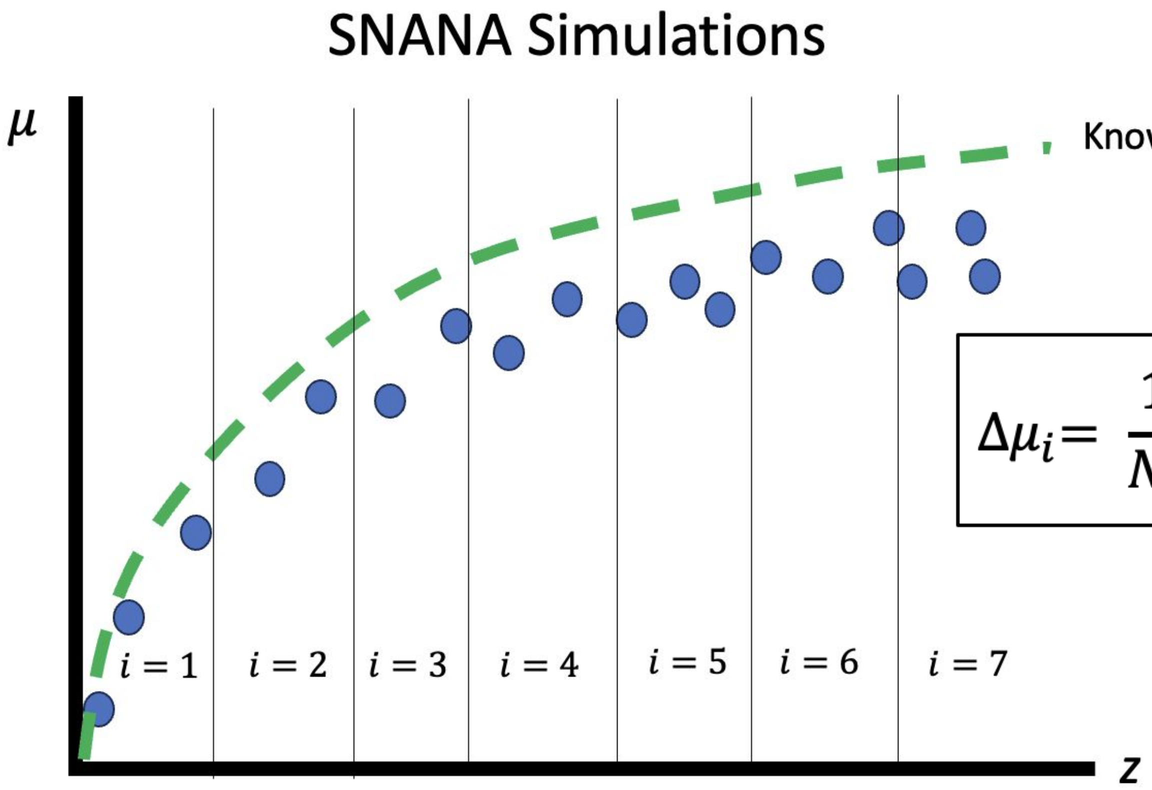




The Impact of Selection Effects

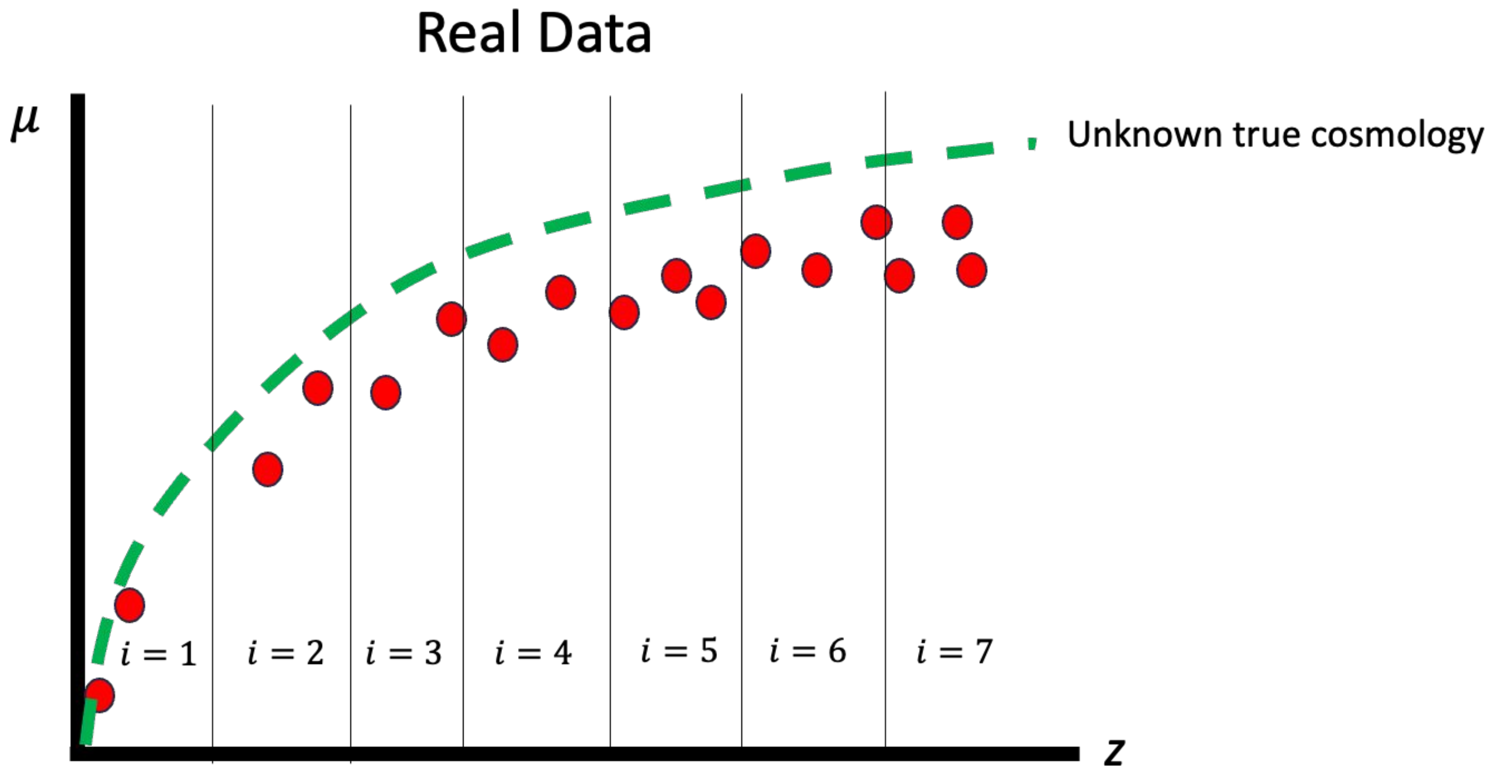


The Impact of Selection Effects

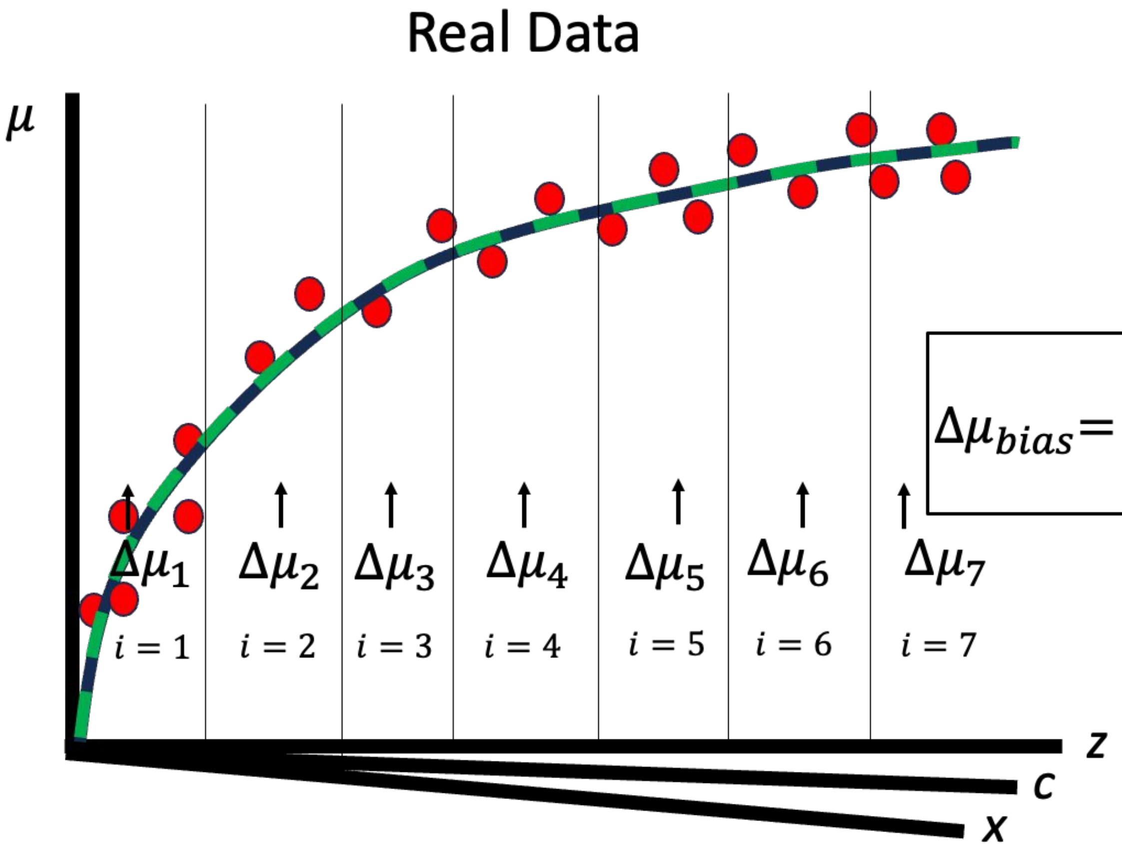


$$\Delta\mu_i = \frac{1}{N_i}\sum_{s=1}^{N_i}(\mu_{s,i}^{fit} - \mu_{s,i}^{true})$$

The Impact of Selection Effects



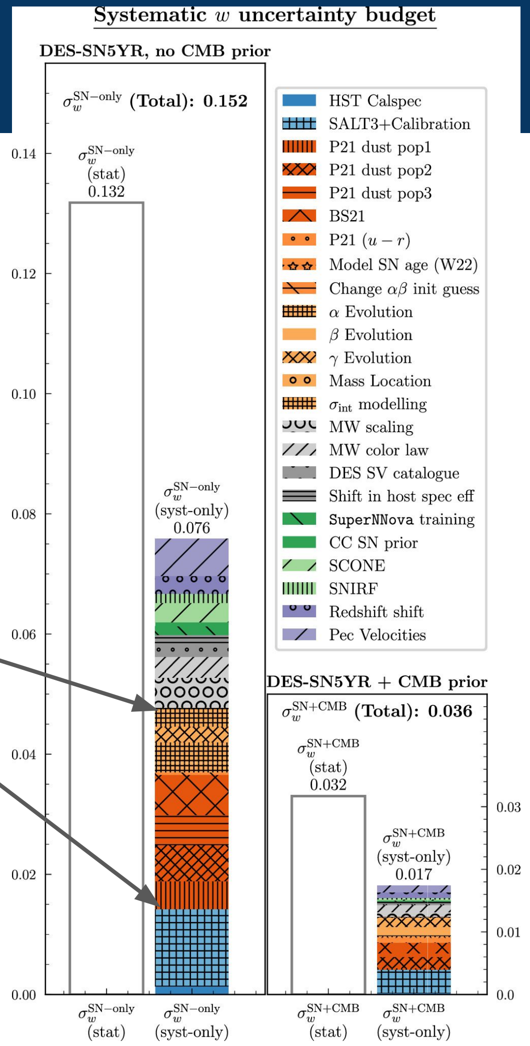
The Impact of Selection Effects



$$\Delta\mu_{bias} = \frac{1}{N_i}\sum_{s=1}^{N_i}(\mu_{s,i}^{fit} - \mu_{s,i}^{true})$$

Challenges in Supernova Cosmology

The majority of systematic uncertainty in DES5YR relates to unknown astrophysics



Vincenzi+24

Supernova cosmology is inseparably linked to supernova astrophysics

Recipe for a Type Ia Supernova Model

Type Ia SN SED

=

+

+

+

Recipe for a Type Ia Supernova Model

Type Ia SN SED

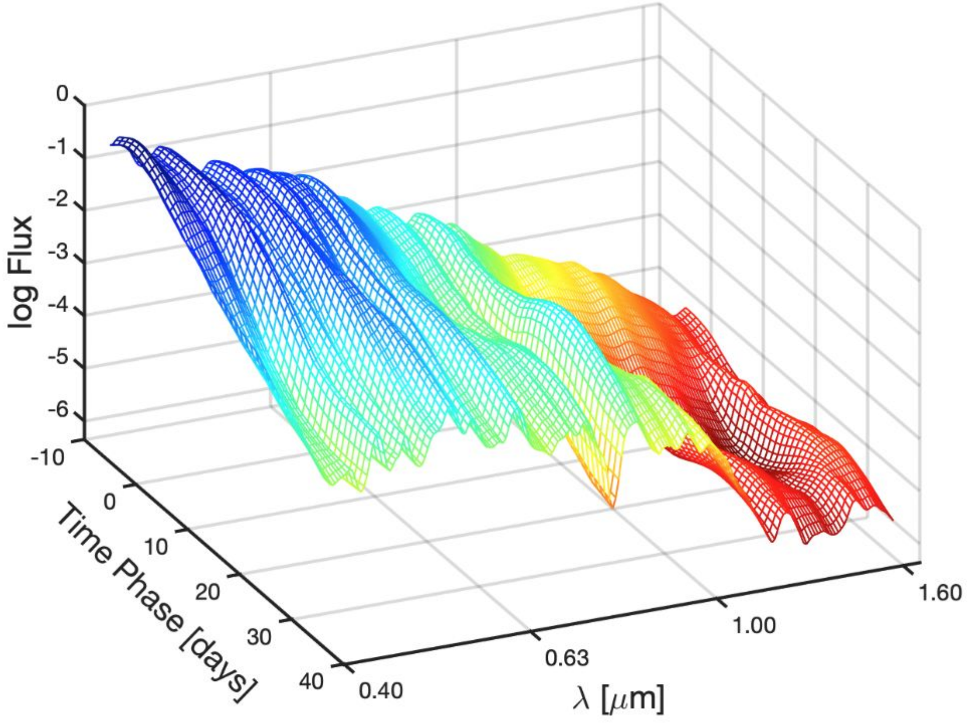
=

Mean Intrinsic SED

+

+

+



Recipe for a Type Ia Supernova Model

Type Ia SN SED

=

Mean Intrinsic SED

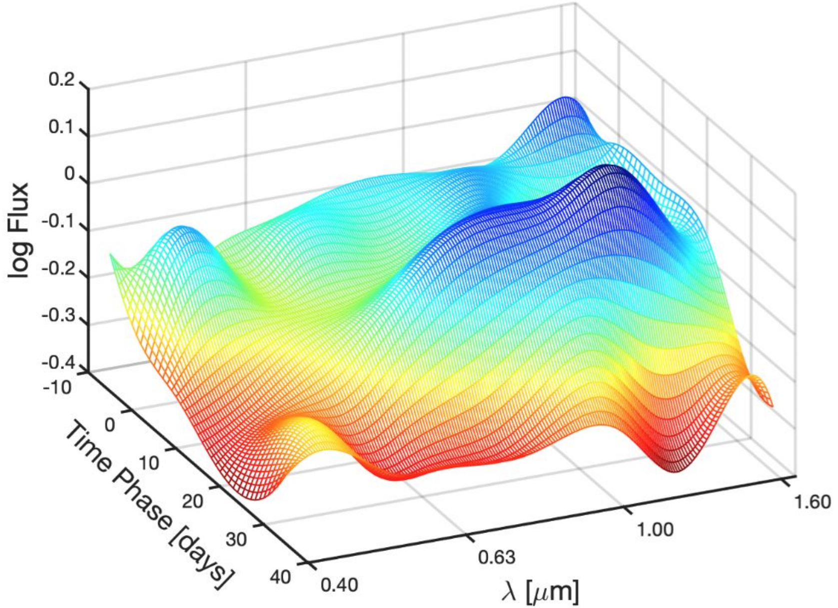
+

Effect of Stretch

+

+

θ_1 ×



Recipe for a Type Ia Supernova Model

Type Ia SN SED

=

Mean Intrinsic SED

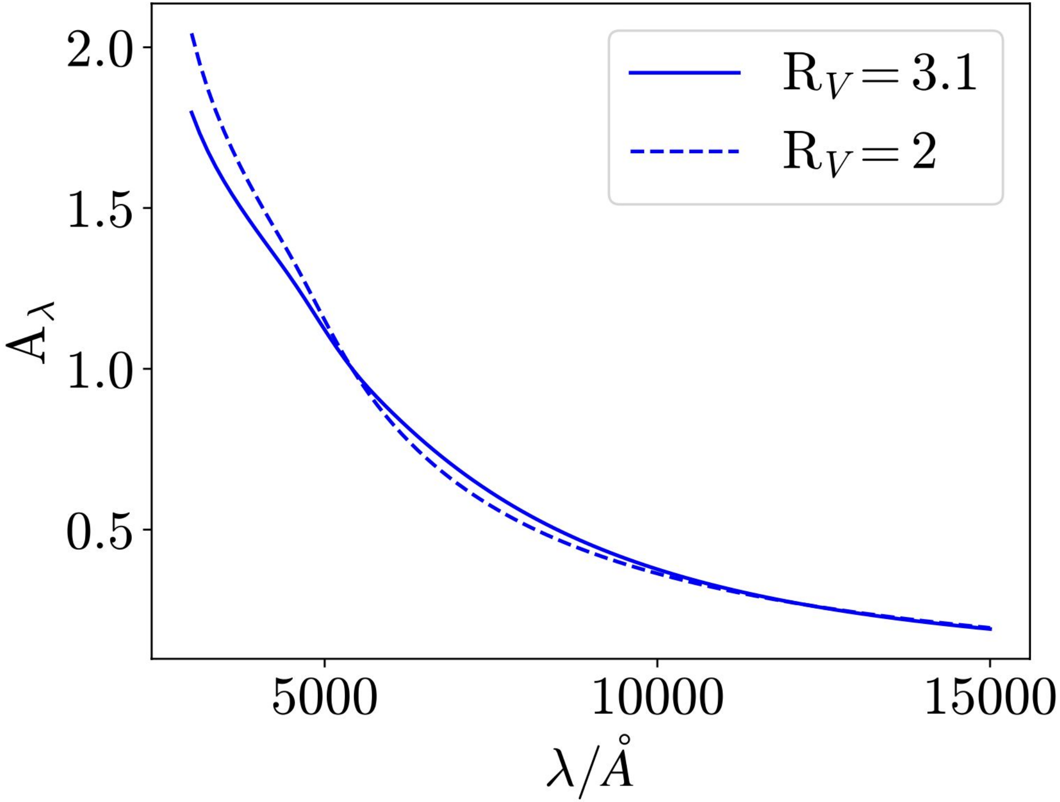
+

Effect of Stretch

+

Dust Extinction

+



Recipe for a Type Ia Supernova Model

Type Ia SN SED

=

Mean Intrinsic SED

+

Effect of Stretch

+

Dust Extinction

+

$$R_V \sim N(\mu_R, \sigma_R^2)$$

$$A_V \sim Exp(\tau_A)$$

Recipe for a Type Ia Supernova Model

Type Ia SN SED

=

Mean Intrinsic SED

+

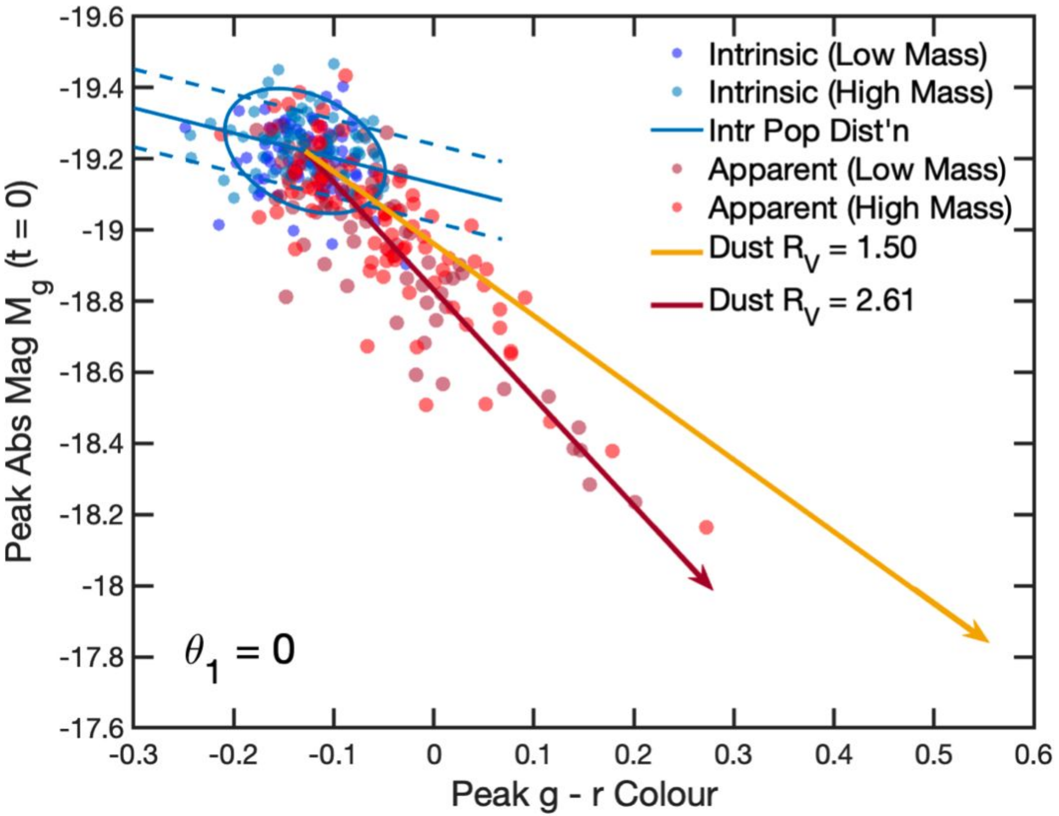
Effect of Stretch

+

Dust Extinction

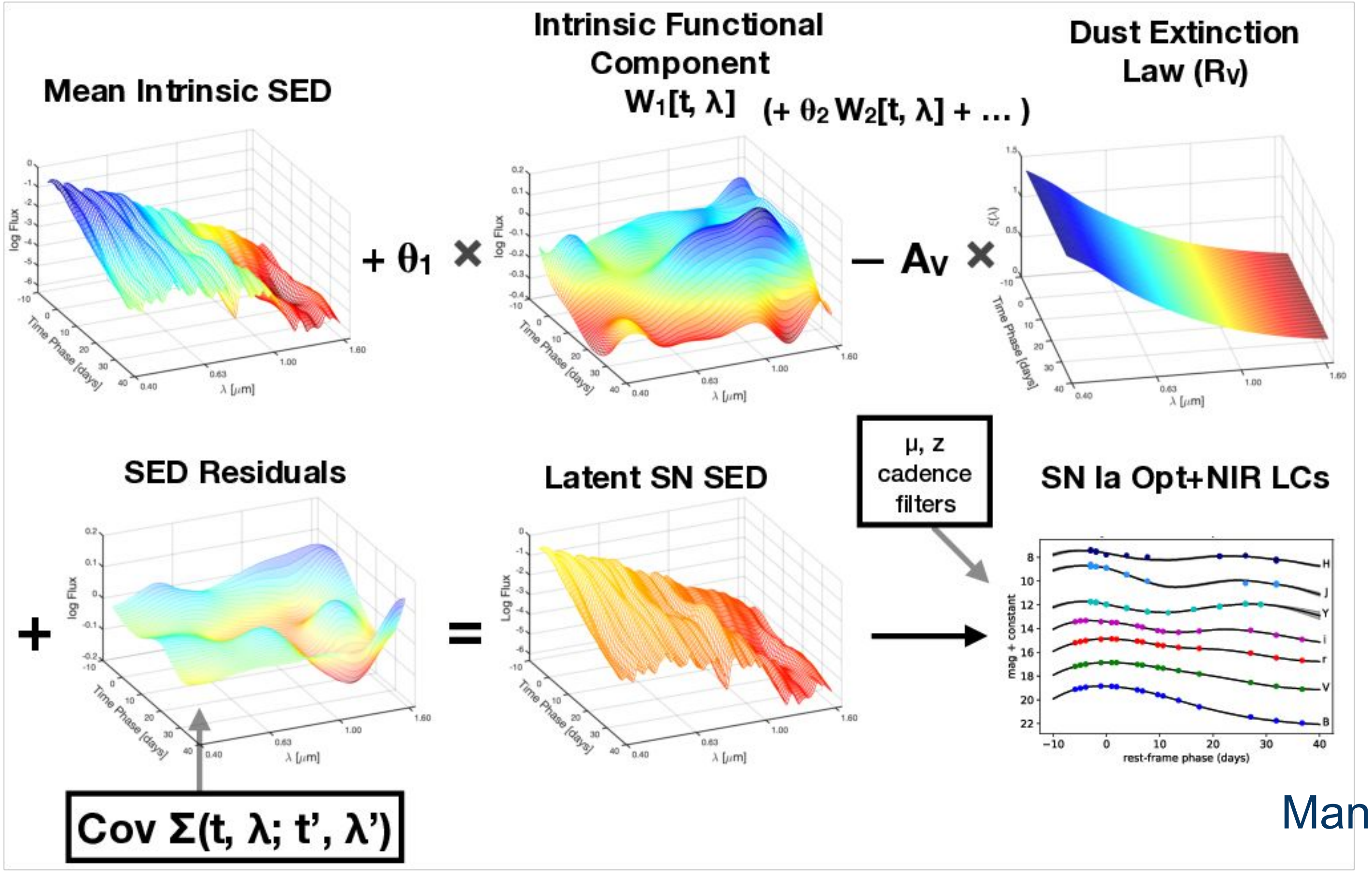
+

Intrinsic Colour



Thorp+2021

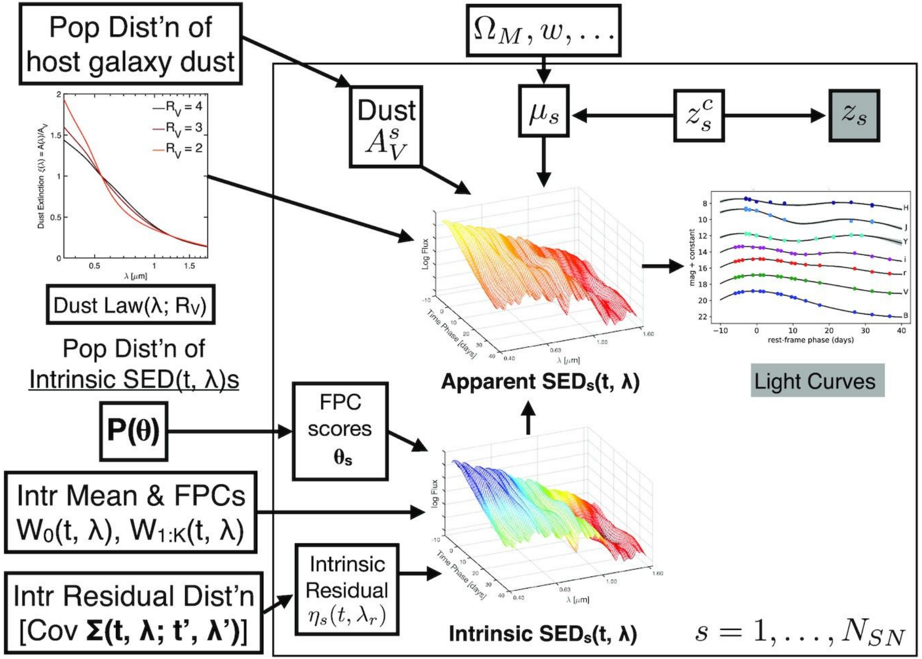
The BayeSN Model



Mandel+2020

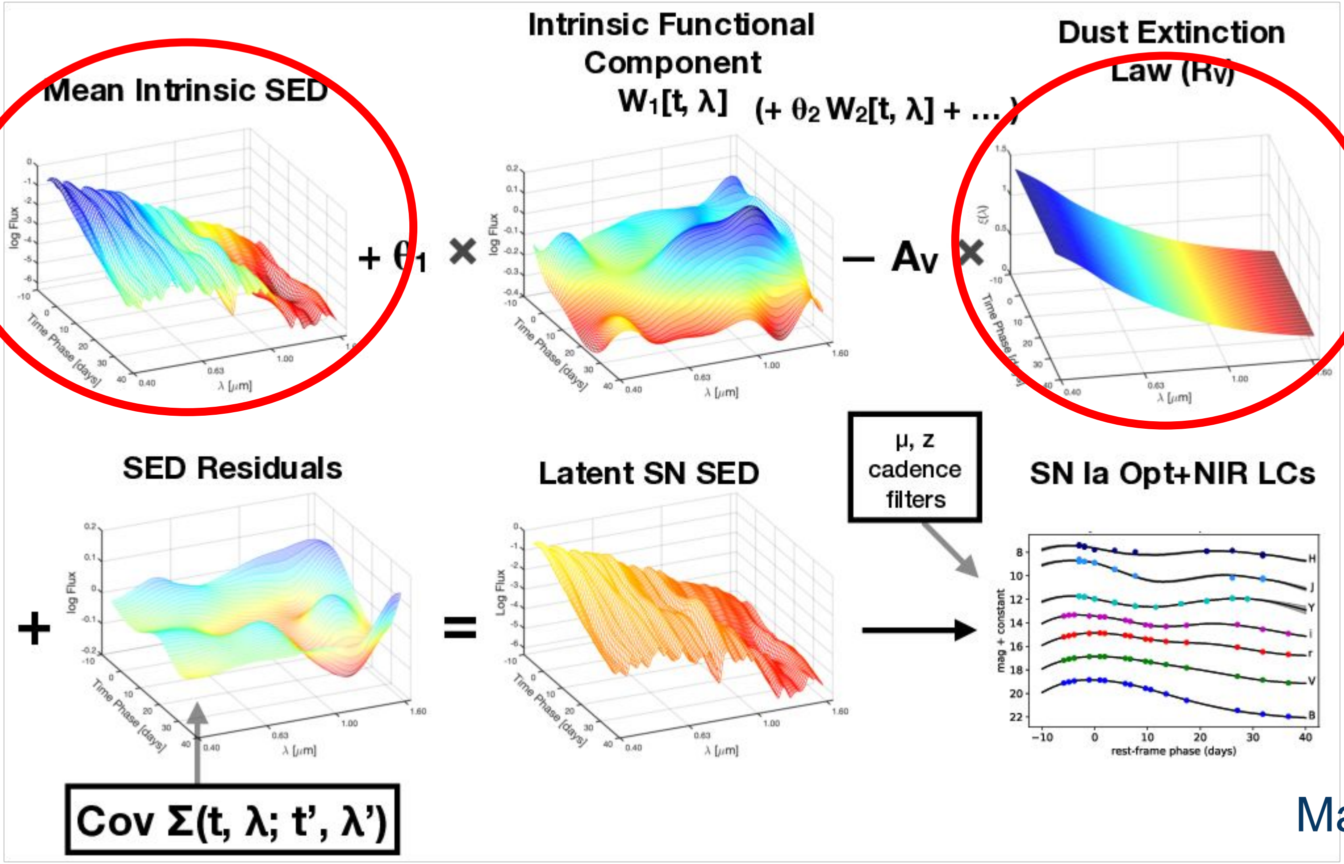
Advantages of BayeSN

- Hierarchical Bayesian Model
  - Allows for separate treatment of dust and intrinsic variation
- Full SED model
  - Uses full light curve in all bands rather than just single colour/magnitude pairs
- Fit for distance jointly with stretch/dust
  - Distance marginalised over dust/intrinsic variation



Mandel+2020

Investigating the Mass Step



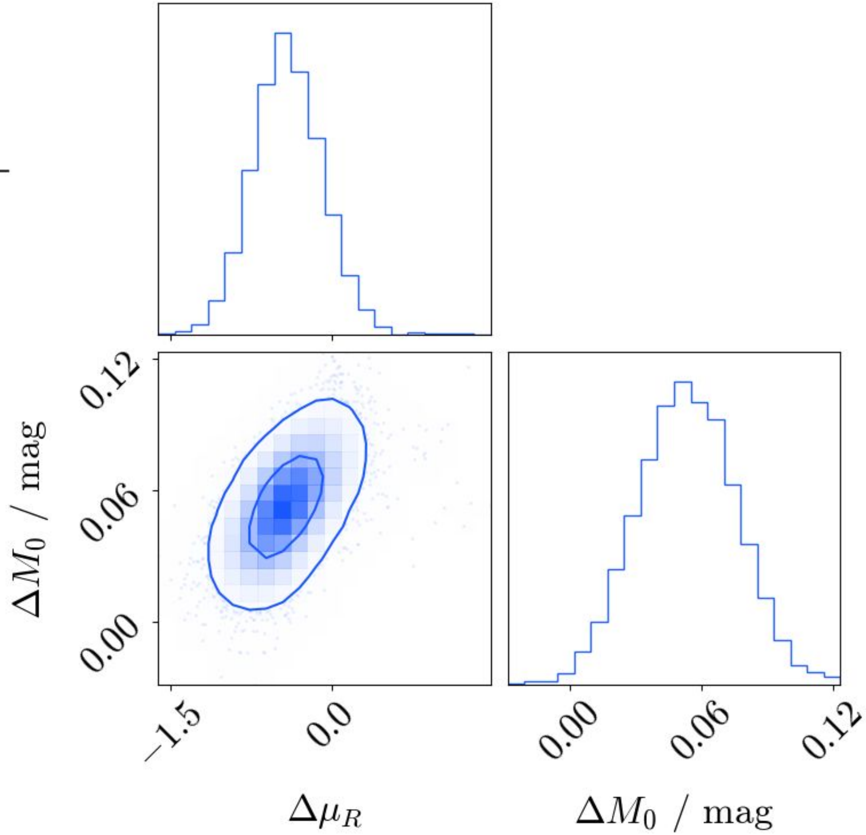
Mandel+2020

Cause of the Mass Step

Sample	Intrinsic Mass Step / mag
Combined*	0.049±0.019
DES5YR	0.053±0.022

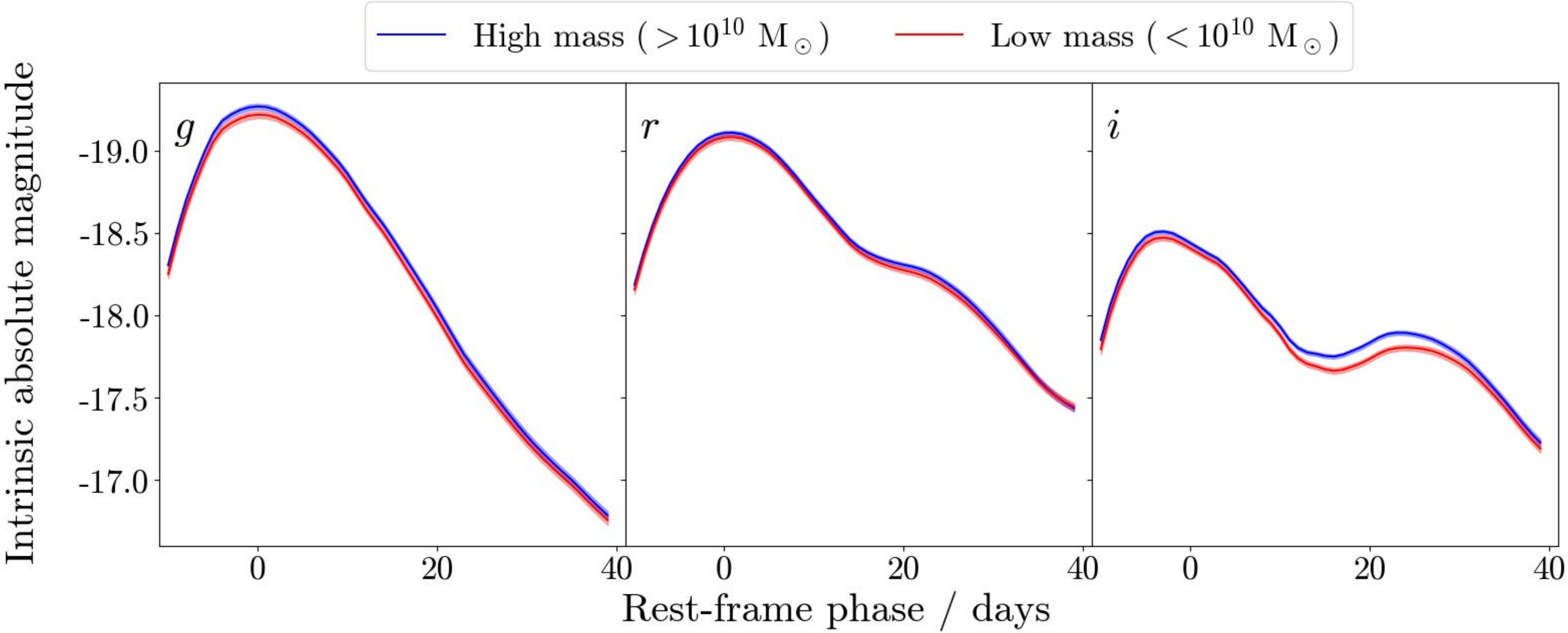
*Combined = DES3YR+Foundation+PS1MD

Dust may play a role, but BayeSN supports an intrinsic contribution to the mass step



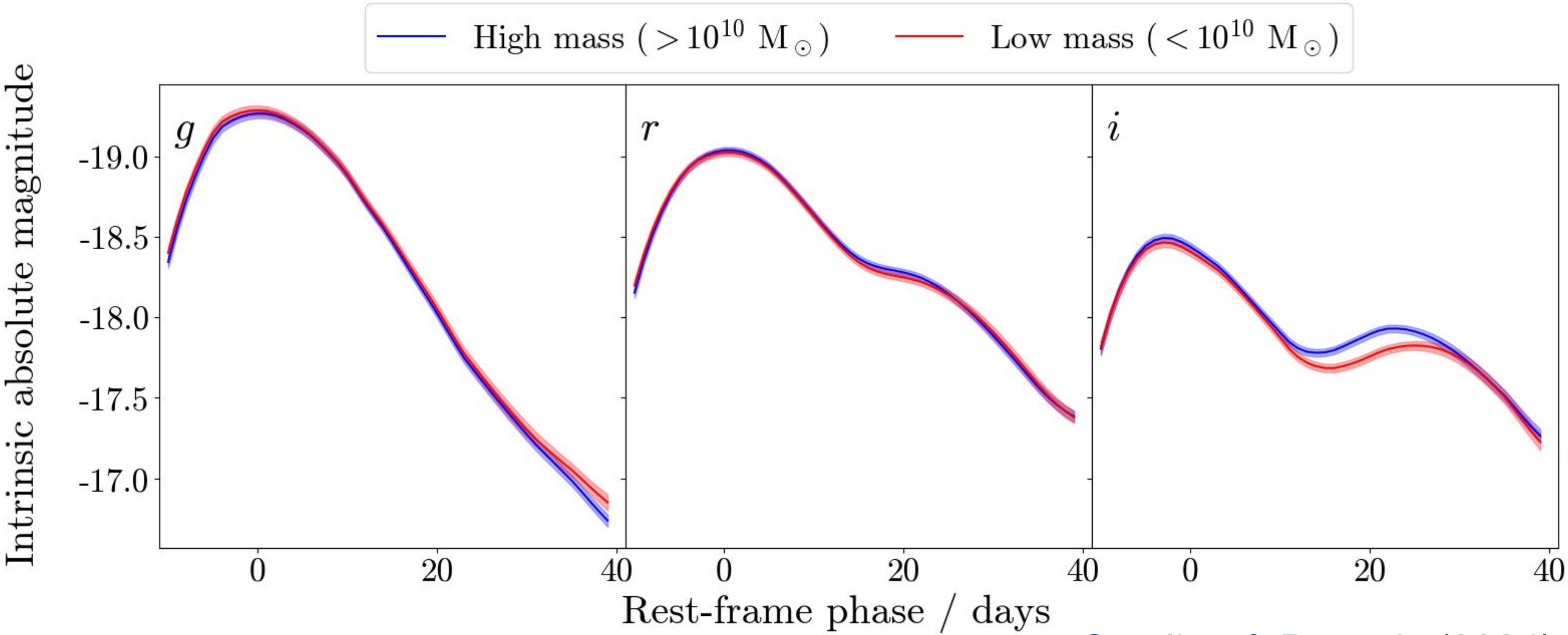
Grayling & Popovic (2024)

DES3YR + Foundation + PS1MD



Grayling+2024

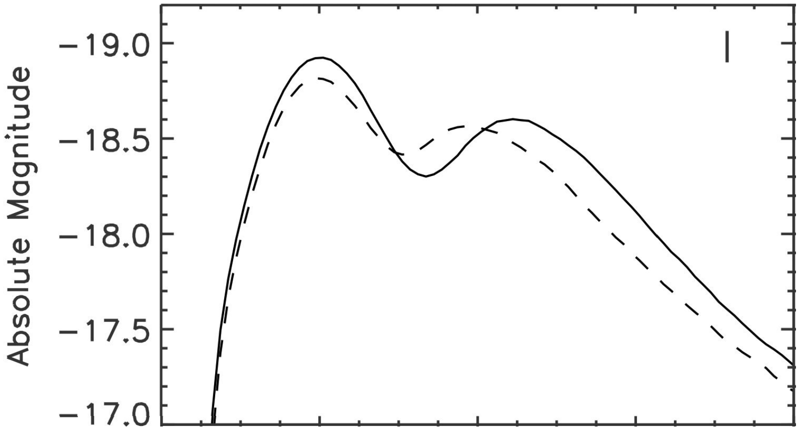
DES5YR



Grayling & Popovic (2024)

What's Causing the i-band Difference?

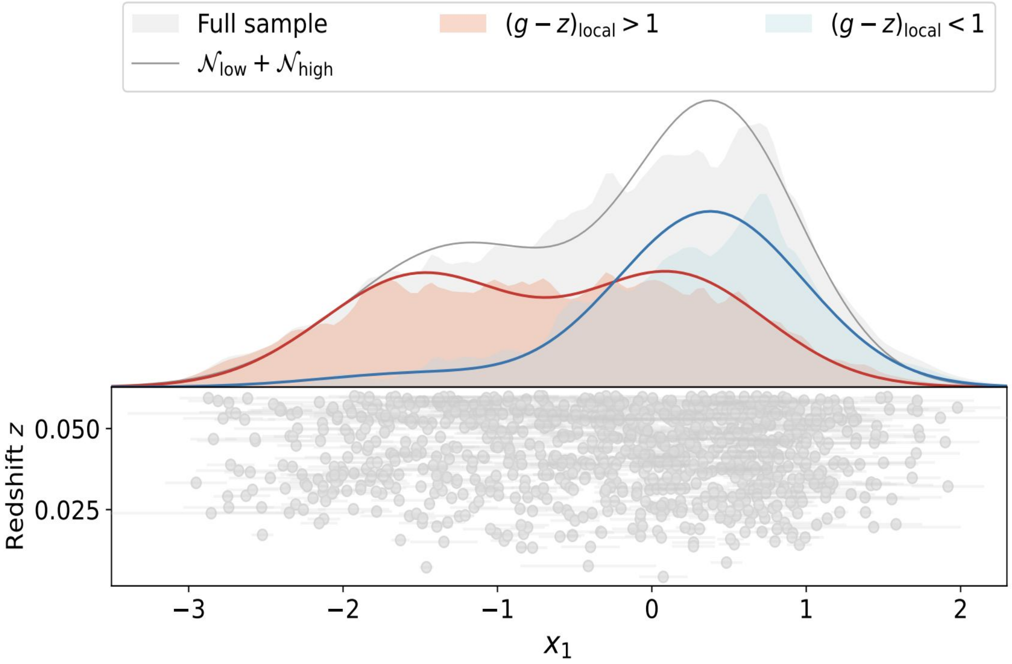
- *Something* is causing an intrinsic difference between SNe Ia in different environments, specifically in i-band.
- Kasen (2006) predicts that metallicity will have stronger impact on secondary maximum than around peak.
- Difference specifically in i-band; Deckers et al. (2024) find differences between behaviour of second maximum in i and r-band.



From Kasen (2006); dashed line = higher metallicity

Understanding the Environmental Dependence

- Mass can't be driving the difference, must be a proxy for something
- We can learn more from looking at the local environments in which SNe Ia explode
- Could be related to progenitor age or metallicity



Ginolin+2025

Could the SN Ia population evolve with redshift?

- The universe evolves with time
- Even if type Ia supernovae have constant properties, what if the dust in their host galaxies changes over time?
- We can approach this problem with BayeSN

$$R_V \sim N(\mu_R, \sigma_R^2)$$

$$\downarrow$$

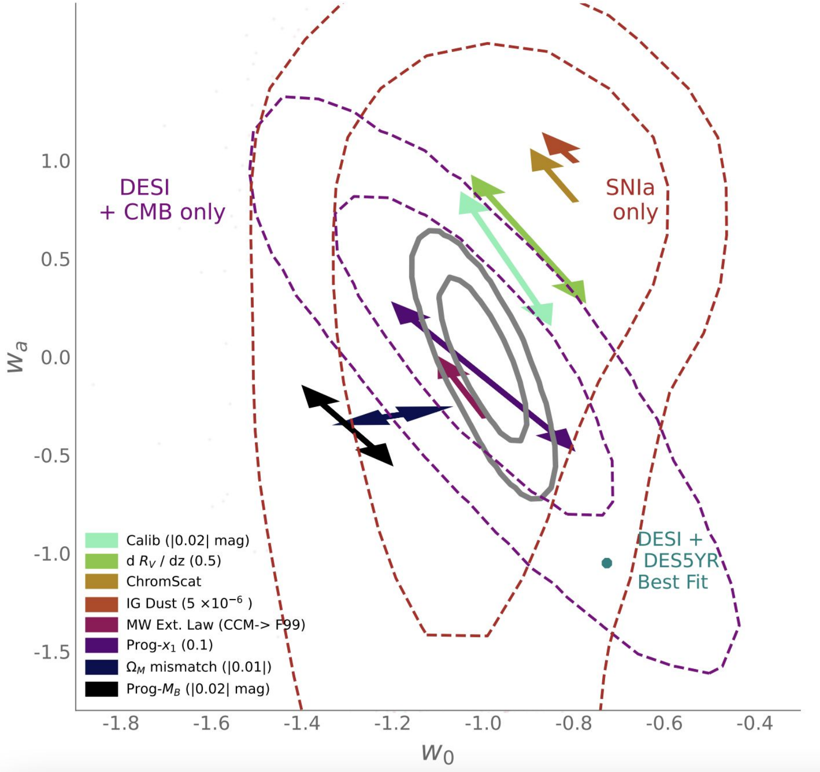
$$R_V \sim N(\mu_R(z), \sigma_R^2)$$

$$\mu_R(z) = \mu_{R,z0} + \eta_R z$$

It's really hard to tell

Parameter	Value
$\mu_{R,z0}$	2.58±0.17
η_R	-0.38±0.70

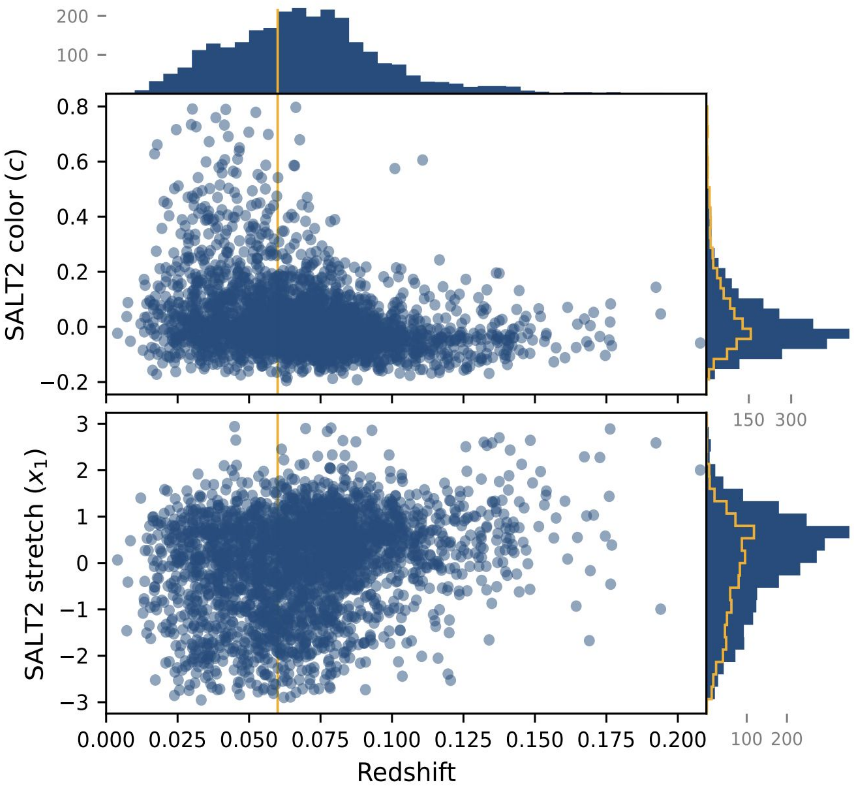
From Grayling+2024, based on 475 type Ia supernovae at z < 0.4



Dhawan+2024

Prospects for probing redshift evolution

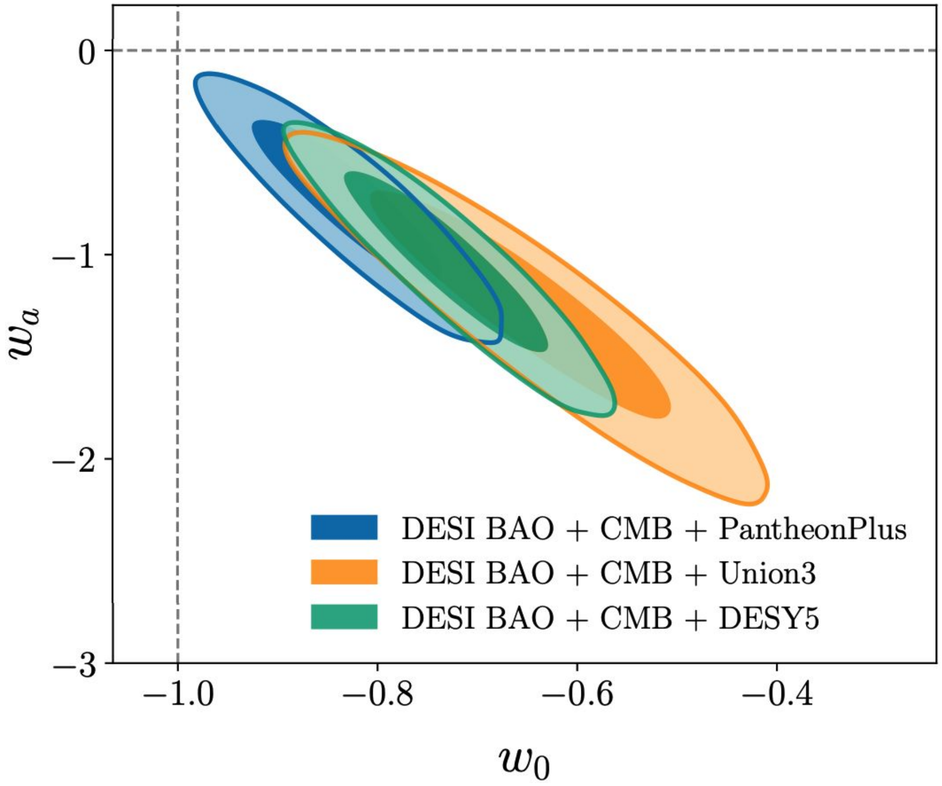
- Disentangling redshift evolution from selection effects is really challenging
- Need good volume limited samples
- An increasing number of samples with good samples minimally impacted by selection effects (e.g. ZTF) are now available



Rigault+24

Implications for SN Cosmology

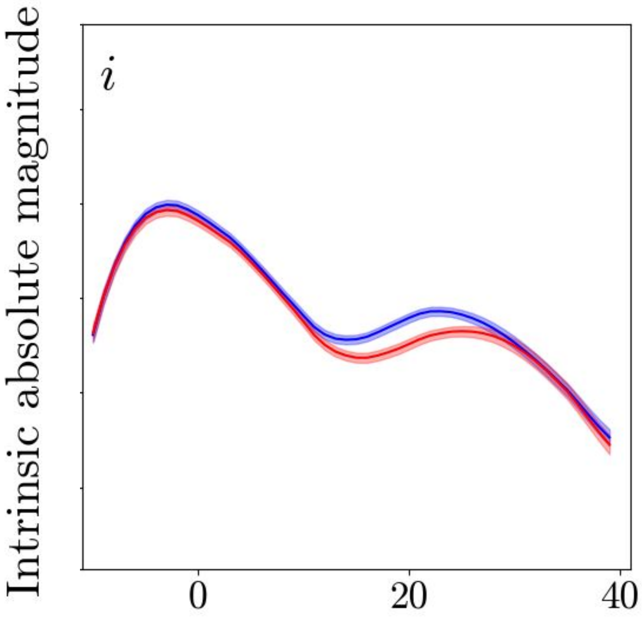
- Supernova cosmology is reliant on simulations - we need to get the forward model right
- We need to uncover exactly what is driving the environmental dependence that we observe
- Also vital to constrain any evolution of dust properties with redshift
- BayeSN can play a vital role in solving these problems



DESI Collaboration 2024

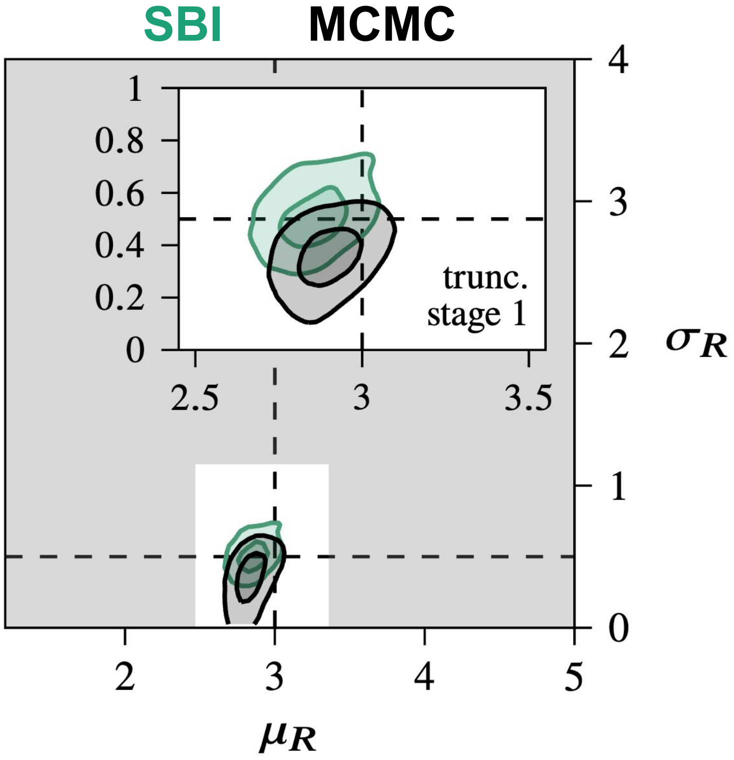
Plans for the Future

- Applying BayeSN to both spectra and photometry
  - Hopefully we can isolate differences in specific spectral features
  - Will provide more physical insight, better understanding of what is driving i-band maximum difference



Plans for the Future

- Cosmology with BayeSN
  - Currently working on plugging BayeSN into ‘standard’ SN cosmology pipeline
  - Developing new, principled methods for doing supernova cosmology using Simulation-based Inference (SBI)
  - SBI is the key to inferring cosmological parameters jointly with properties of SNe Ia



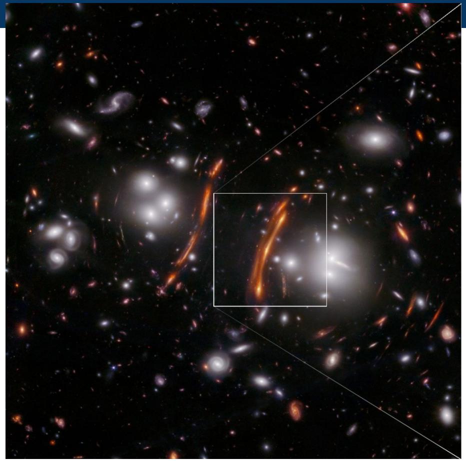
Karchev, Grayling et al. (2024)

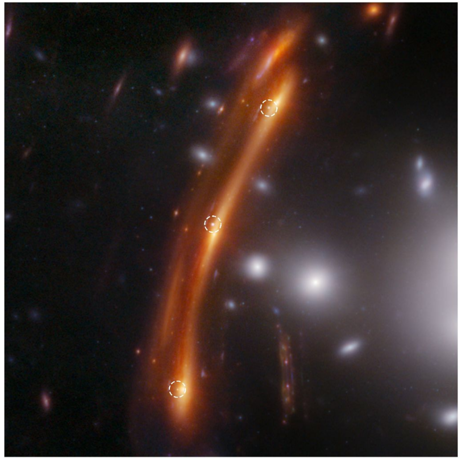
BayeSN-TD:

Adapting BayeSN for strongly-lensed SNe

Why strongly lensed supernovae?

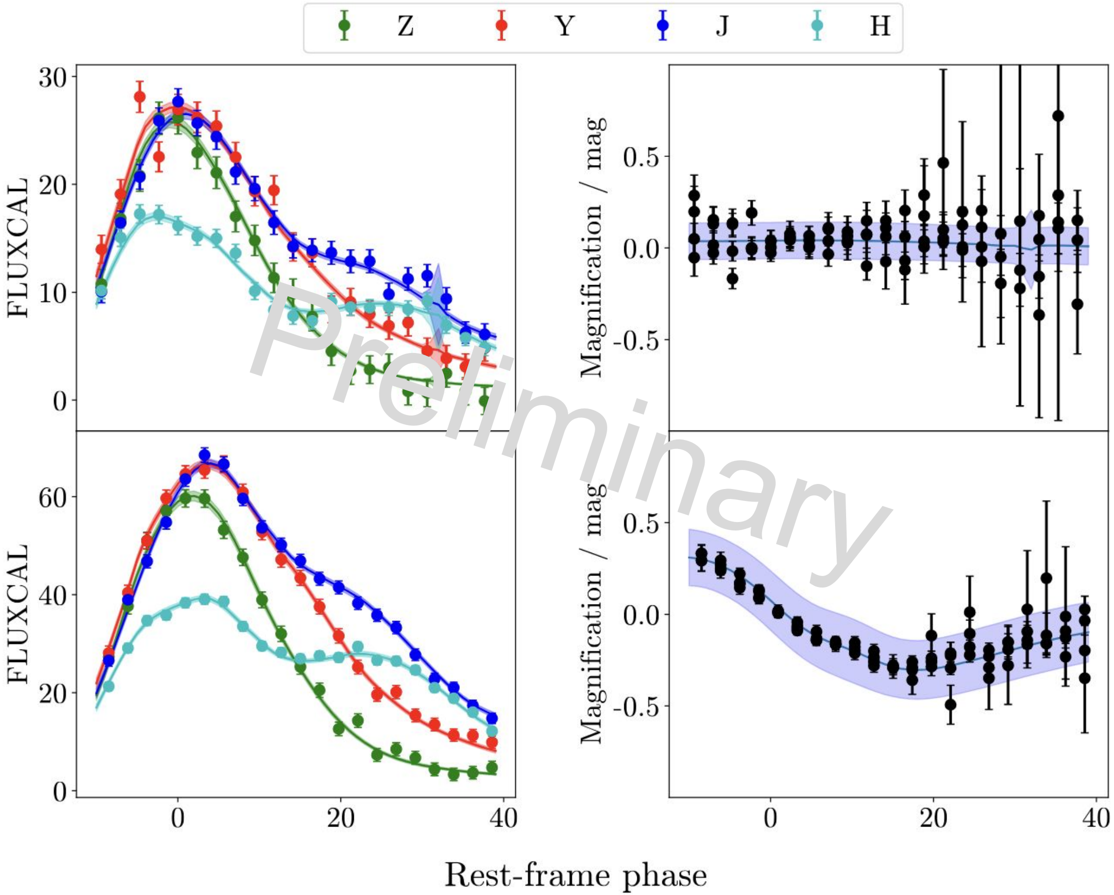
- Independent probe of Hubble constant
- Microlensing caused problems
    - Time-varying magnification during SN light curve, causes deviations from our SED templates
- We have adapted BayeSN to fit multiply-imaged strongly-lensed supernovae, including a Gaussian Process treatment for microlensing





Example on simulation

We can constrain microlensing deviations successfully!



Summary

- With BayeSN, we find evidence to support an intrinsic component of the mass step though dust may also play a role
- We find a significant environmental-dependence of the intrinsic properties of SNe Ia, specifically on the i-band secondary maximum
- Current data cannot yet constrain supernova host galaxy dust evolution with redshift
- Lots of ongoing work for BayeSN e.g. training on spectra, moving towards cosmology

MNRAS paper

New arxiv paper